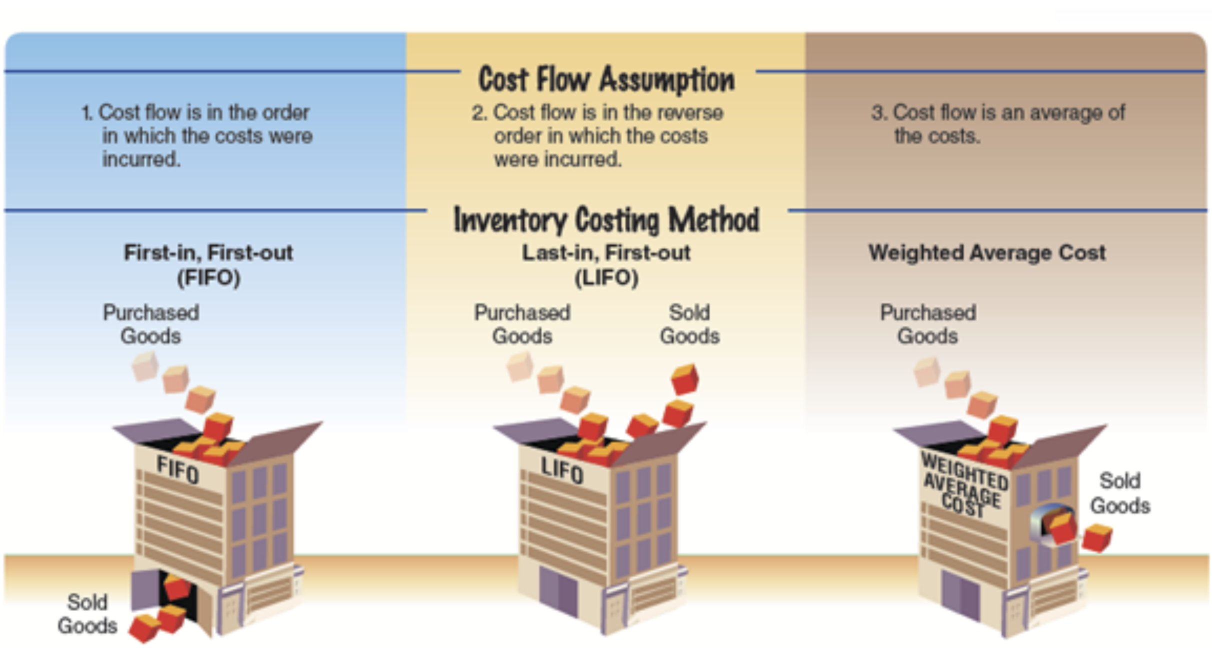
6-2 Inventory Cost Flow Assumptions

Objective 2 - Describe three inventory cost flow assumptions and how they impact the income statement and balance sheet.

An accounting issue arises when identical units of merchandise are acquired at different unit costs during a period. In such cases, when an item is sold, it is necessary to determine its cost using a cost flow assumption and related inventory costing method. Three common cost flow assumptions and related inventory costing methods are shown in Exhibit 1.

Exhibit 1

Cost Flow Assumptions



To illustrate, assume that three identical units of merchandise are purchased during May, as follows:

			Units	Cost
May	10	Purchase	1	$ 9
	18	Purchase	1	13
	24	Purchase	1	14
Total			3	$36

Average cost per unit: $12 ($36 ÷ 3 units)

Assume that one unit is sold on May 30 for $20. Depending upon which unit was sold, the gross profit varies from $11 to $6, computed as follows:

	May 10 Unit Sold	May 18 Unit Sold	May 24 Unit Sold
Sales	$20	$ 20	$ 20
Cost of goods sold	(9)	(13)	(14)
Gross profit	$11	$ 7	$ 6
Ending inventory	$27	$ 23	$ 22
	($13 + $14)	($9 + $14)	($9 + $13)

Under the **specific identification inventory cost flow method** (The method of inventory costing in which a unit sold is identified with a specific purchase.), the unit sold is identified with a specific purchase. The ending inventory is made up of the remaining units on hand. Thus, the gross profit, cost of goods sold, and ending inventory can vary as illustrated. For example, if the May 18 unit was sold, the cost of goods sold is $13, the gross profit is $7, and the ending inventory is $23.

The specific identification method is not practical unless each inventory unit can be separately identified. For example, an automobile dealer may use the specific identification method because each automobile has a unique serial number. However, most businesses cannot identify each inventory unit separately. In such cases, one of the following three inventory cost flow methods is used.

Under the **first-in, first-out (FIFO) inventory cost flow method** (The method of inventory costing based on the assumption that the first units purchased are the first units sold; the method of inventory costing based on the assumption that the costs of merchandise sold should be charged against revenue in the order in which the costs were incurred.), the first units purchased are assumed to be sold and the ending inventory is made up of the most recent purchases. In the preceding example, the May 10 unit would be assumed to have been sold. Thus, the gross profit would be $11, and the ending inventory would be $27 ($13 + $14).

Under the **last-in, first-out (LIFO) inventory cost flow method** (A method of inventory costing based on the assumption that the last units purchased are assumed to be sold and the ending inventory is made up of the first purchases.), the last units purchased are assumed to be sold and the ending inventory is made up of the first purchases. In the preceding example, the May 24 unit would be assumed to have been sold. Thus, the gross profit would be $6, and the ending inventory would be $22 ($9 + $13).

Under the **weighted average inventory cost flow method** (A method of inventory costing in which the cost of the units sold and in ending inventory is a weighted average of the purchase costs.), sometimes called the *average cost flow method*, the cost of the units sold and in ending inventory is a weighted average of the purchase costs. The purchase costs are weighted by the quantities purchased at each cost, thus the term *weighted average*. In the preceding example, the cost of the unit sold would be $12 ($36 ÷ 3 units), the gross profit would be $8 ($20 − $12), and the ending inventory would be $24 ($12 × 2 units). In this example, the purchase costs are weighted equally, since the same quantity (one) was purchased at each cost.

Business Insight

RFID and Specific Identification

Companies like **Best Buy Co., Inc. (BBY)** have adopted the use of radio frequency identification (RFID) tags as a means to track and control inventory. Similar to a bar code, an RFID tag stores large amounts of data, including its maker/supplier and cost. The item's data are transmitted by radio waves to a reader instead of by a scan of a bar code. RFID tags have the potential to allow the use of the specific identification method for valuing most merchandise. However, the high cost of RFID tags has limited its widespread use among all but the largest retailers.

Link to Best Buy

Best Buy uses the weighted average cost method for its inventory.

The three inventory cost flow methods, FIFO, LIFO, and weighted average, are shown in Exhibit 2.

Exhibit 2

Inventory Costing Methods